

Dutch Fleet and Electrification Brief

NorthLedger Insights - automated analysis

DUTCH FLEET AND ELECTRIFICATION BRIEF

RDW register extract: 16,852,477 vehicles, 16,983,510 fuel records

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1. SCOPE AND METHOD

Source tables: rdw_vehicles (16,852,477 rows) and rdw_fuel (16,983,510 rows), joined on kenteken. Fuel mix is computed with brandstof_volgnummer = 1 as the primary fuel, with multi-fuel (hybrid) vehicles counted explicitly and separately. 15,145,752 vehicles (89.9% of the register) have at least one fuel record; 1,706,725 do not, and 1,703,949 of those (99.8%) are non-motorized units (trailers, semi-trailers, agricultural implements). Only 2,080 passenger cars lack a fuel record. First-registration date is not present in this extract, so vervaldatum_apk (APK expiry, YYYYMMDD strings) is used as the recency signal.

2. TABLE PROFILE

rdw_vehicles: 16,852,477 rows, 16,852,477 distinct kenteken (join key is unique, no duplicate plates), 15 distinct voertuigsoort values, 11,447 distinct merk strings. Data quality: 234 null-like brand values, 43,967 empty handelsbenaming; brand casing is already consistent (normalizing case and padding changed no counts). vervaldatum_apk: 12,246,703 valid 8-character dates, 4,605,774 blank (27.3% of the register), spanning years 1988 to 2031 (a few thousand rows in the oldest and newest years look like artifacts).

rdw_fuel: 16,983,510 rows over 15,149,149 distinct kenteken, 9 distinct fuel descriptions (including 1 blank record), brandstof_volgnummer from 1 to 3. 1,832,296 kenteken carry more than one fuel row; 2,065 carry three. Row arithmetic reconciles exactly: 15,149,149 kenteken plus 1,834,361 extra fuel rows equals 16,983,510.

3. FUEL MIX (primary fuel, volgnummer = 1; n = 15,145,752 vehicles with fuel record)

Table: fuel type | count | share

Benzine	10,057,313	66.4%
Diesel	2,614,491	17.3%
Elektriciteit	2,400,781	15.9%
LPG	56,657	0.37%
CNG	10,513	0.07%
Alcohol	3,963	0.03%
LNG	1,544	0.01%
Waterstof	489	0.003%
(blank)	1	0.000%

Of the full register of 16,852,477 vehicles, 2,400,781 (14.2%) list electricity as primary fuel.

4. HYBRIDS AND MULTI-FUEL VEHICLES

1,832,296 kenteken (12.1% of the 15,145,752 fuel-record vehicles) have more than one fuel type. Largest combinations: Elektriciteit + Benzine 1,654,235 (1,188,653 with electricity listed first, 465,582 with benzine first); Benzine + LPG 140,943 (dual-fuel, not plug-in); Diesel + Elektriciteit 25,398; the rest are small volumes.

Formal electrification split across all 15,149,149 fuel-record kenteken:

Table: category | count | share

- BEV, electricity only:	1,205,640	7.96%
- PHEV (electricity + benzine/diesel):	1,680,533	11.09%
- Electricity + other fuel:	1,228	0.01%

- No electricity at all: 12,261,748 80.94%

So roughly 1 in 5 fuel-record vehicles (19.1%) has an electric drivetrain component.

5. PASSENGER CAR SEGMENT (n = 10,802,463 cars with fuel record)

Benzine 7,874,871 (72.9%); Elektriciteit 2,085,122 (19.3%); Diesel 794,417 (7.4%); LPG 37,613; CNG 6,184; Alcohol 3,882; Waterstof 371; LNG 3.

Electrified: BEV 892,313 (8.3%), PHEV 1,672,577 (15.5%), total any-electric 2,564,854 (23.7% of cars).

6. FLEET RECENCY BY APK EXPIRY (all 16,852,477 vehicles)

Table: status | count | share of register

- APK valid into 2027 or later: 9,253,924 54.9% of register (75.6% of the 12,246,703 dated)
- APK valid 2025-2026: 2,668,465 15.8% of register (21.8% of dated)
- APK expired 2024 or earlier: 324,314 1.9% of register (2.6% of dated)
- No APK date: 4,605,774 27.3% of register

The blank group is dominated by vehicle classes without periodic inspection: bromfiets (mopeds) 1,372,910; motorfiets 898,906; aanhangwagen 757,717; middenasaanhangwagen 428,276; tractors and implements 357,664 plus 269,480. Only 263,117 passenger cars have no APK date (likely exported or suspended registrations, plus very recent new registrations). The small expired tail (324,314) reflects register hygiene: vehicles that lapse are typically exported or deregistered and drop out.

7. TOP 10 VEHICLE TYPES (of 15 types, by count)

Table: type | count | share of 16,852,477

1. Personenauto 10,804,533 (64.1%)
2. Bedrijfsauto 1,503,089 (8.9%)
3. Bromfiets 1,372,910 (8.1%)
4. Motorfiets 898,906 (5.3%)
5. Aanhangwagen 769,527 (4.6%)
6. Middenasaanhangwagen 435,030 (2.6%)
7. Land- of bosbouwtrekker 357,674 (2.1%)
8. Land- of bosb aanhw of getr uitr stuk 269,480 (1.6%)
9. Oplegger 220,394 (1.3%)
10. Motorrijtuig met beperkte snelheid 113,581 (0.7%)

8. TOP 10 BRANDS (by count, all vehicle types)

Table: brand | count

1. VOLKSWAGEN 1,524,531
2. PEUGEOT 844,003
3. TOYOTA 825,562
4. FORD 810,241
5. RENAULT 805,787
6. OPEL 741,991
7. MERCEDES-BENZ 717,412
8. BMW 625,168
9. KIA 512,393
10. VOLVO 506,987

9. ELECTRIFICATION BY APK COHORT (the transition, quantified)

All vehicles, primary fuel = electricity:

- APK valid 2025 or later: 2,117,374 of 11,922,389 = 17.76% (any-electric: 2,597,059 = 21.78%)
- APK expired before 2023: 502 of 218,107 = 0.23% (any-electric: 887 = 0.41%)

The recent cohort is roughly 77 times more electric by primary-fuel share than the pre-2023 cohort.

Passenger cars only:

- APK 2025 or later: 2,035,794 of 10,289,167 = 19.79%
- APK 2023-2024: 932 of 76,869 = 1.21%
- APK before 2023: 263 of 175,390 = 0.15%

That is a 132x gap between the newest active cohort and the pre-2023 cohort.

Passenger-car primary-EV share by APK expiry year (table): year | EV | cars | share

2025: 14,007 of 252,659 = 5.5%

2026: 179,421 of 1,931,500 = 9.3%

2027: 662,670 of 6,177,077 = 10.7%

2028: 599,739 of 1,243,519 = 48.2%

2029: 355,359 of 435,734 = 81.5%

2030: 224,598 of 247,002 = 91.0%

Among cars with APK in 2029 or later, 579,926 of the electric cars carry emissieklasse Z (zero emission) versus 99,899 benzine cars at class 6, confirming these are the newest registrations. Because a new Dutch passenger car typically faces its first inspection about 4 years after first registration, the 2028-2030 APK cohorts correspond to cars first registered roughly 2024-2026. Best estimate with honest uncertainty: new-car registrations are currently running at an EV share in the 48% to 91% band depending on the cohort cut, most likely 60-80% for the very newest cars. This is a proxy estimate, not a registration-date measurement.

10. LIMITATIONS

1. No first-registration date exists in this extract; `vervaldatum_apk` is a proxy for recency. It is blank for 27.3% of the register (mostly mopeds, motorcycles and trailers), so recency analysis is reliable mainly for cars and vans.
2. The expired-APK tail (324,314) is censored: expired vehicles are usually exported or deregistered and leave the register, so the true number of lapsed vehicles is understated.
3. Fuel mix covers the 15,145,752 vehicles that have a fuel record; 1,706,725 vehicles are excluded, of which 99.8% are correctly excluded non-motorized trailers.
4. `volgnummer = 1` is assumed to be the primary fuel; PHEV ordering is inconsistent in the source (1,188,653 list electricity first, 465,582 list benzine first), so primary-fuel shares are reported alongside any-electricity shares to avoid over- or under-counting EVs.
5. Hybrid identification from fuel-row combinations is a proxy; 2,065 kenteken with three fuel rows and rare fuels (alcohol, hydrogen, LNG) are grouped into small residual categories, and one fuel row has a blank description.
6. Outlier APK years (1988, 2031), 43,967 empty trade names and 234 null-like brands exist but are immaterial at register scale.
7. No mileage, ownership, CO2, price or forecast modeling was possible with these fields; `catalogusprijs` and `CO2` fields were not analyzed for this brief. The 48-91% new-registration EV band is a proxy inference, not a measured registration statistic.

11. WHAT A FLEET OPERATOR SHOULD DO

- Size charging and grid capacity for a majority-electric new-car market: the newest registration cohorts are 48-91% electric (best estimate 60-80%).
- Treat PHEVs as a separate asset class: 1.68M plug-in hybrids nationally (1.67M of them cars) are aging out of tax-driven registrations; model their residual values and fuel-cost exposure separately from BEVs.
- Respect the stock-versus-flow gap: benzine still powers 66.4% of all fuel-record vehicles and 72.9% of cars; combustion vehicles will dominate the operating fleet for years even as new registrations flip.
- Note the laggards: `bedrijfsauto` (vans) hold only 38,230 hybrids against 1.5M units, and diesel still leads heavier segments; expect policy pressure and late-stage conversion there.
- Use APK expiry year as a free recency field for targeting: cars with APK due 2028-2030 are the recent registrations.